Chapter 3

Problem 3.1 A cantilever AB of length *l* is subjected to a downward load P at its free end and an upward load P at a distance a from the free end. Draw S.F. and B.M. diagrams for the cantilever.

Solution:

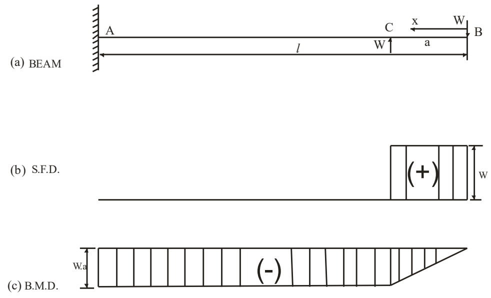

Fig. 3.16

S.F.D.

Section in BC = +W

Section in CA = 0

B.M. at any section in BC = - W.x (Linear);

At B x = 0 $\therefore M_B = 0$

At C x = a $\therefore M_C = -Wa$

B.M. at any section in CA = - W.x + W. (x-a)

= -Wa (Constant)

At C x = a $\therefore M_C = -Wa$

At A x = *l* $\therefore M_A = -Wa$

From C to D the

Problem 3.2 A cantilever beam 5 m long carries concentrated loads of 30, 30 and 30 kN at distance of 1.5, 3 and 4.5 meters from the fixed end. In addition to this the beam carries a uniformly distributed load of 10 kN/m over the entire length of the beam. Draw the S.F. and B.M. diagrams.

Solution :

The loaded beam is shown in Fig. 3.17

S.F. at B = 0

S.F. at C = (10 × 0.5 + 30) = 35 kN

S.F. at D = (10 × 2 + 30 + 30) = 80 kN

S.F. at E=(10 × 3.5 + 30 + 30 + 30)=125 kN

S.F. at A = (10 × 5 + 30 + 30+ 30) = 140 kN

The S.F. diagram is shown in Fig. 3.17(b)

B.M. at B = 0

B.M. at C = $- 10 \times 0.5 \times \frac{0.5}{2} = - 1.25$ kN.m

B.M. at D = $- 10 \times 2 \times \frac{2}{2} - 30 \times 1.5 = - 65$ kN.m

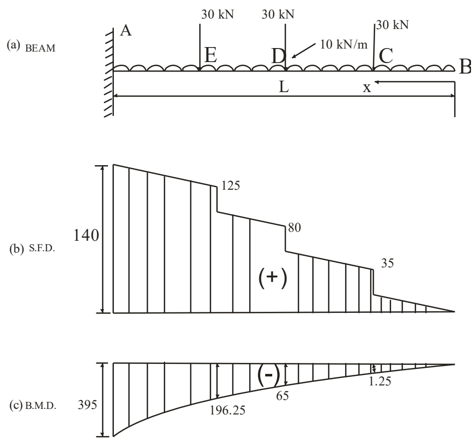

Fig. 3.17

B.M. at E = $- 10 \times 3.5 \times \frac{3.5}{2} - 30 \times 3 - 30 \times 1.5 = - 196.25$ kN.m

B.M. at A = $- 10 \times 5 \times \frac{5}{2} - 30 \times 4.5 - 30 \times 3 - 30 \times 1.5$

$= - 395$ kN.m

The B.M. diagram is shown in Fig. 3.17(c)

Problem 3.3 Draw the shear force and bending moment diagrams for the beam shown in Fig. 3.18. Also find the position and magnitude of the maximum bending moment.
Solution: Let the salient point of the beam be named as shown in the Fig. 3.18. Let R_a and R_b be the vertical reactions at the left and right supports respectively.
Taking moments about the left support, we have

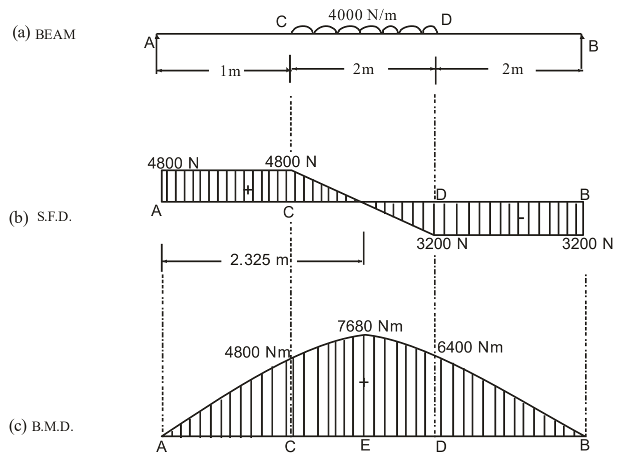

Fig. 3.18

$$R_b \times 5 = 4000 \times 2(1+1)$$

$\therefore$ $R_b = 3200N$

$\therefore$ $R_a = (4000 \times 2) - 3200$

$= 4800N$

S.F. between A and C = +4800N

S.F. between D and B = - 3200 N

Let the S.F. be zero at a section E, x meters from A
Equating the S.F. at E to zero, we have

$$4800 - 4000\ (x - 1) = 0$$

$\therefore$ $x = 2.2$ m

B.M. calculations

B.M. at A = $M_a = 0$

B.M. at C = $M_c = 4800 \times 1 = 4800$ Nm

B.M. at D = $M_a = 3200 \times 2 = 6400$ Nm

B.M. at 2.2 m from A

$$= M_E = 4800 \times 2.2\text{-}4000 \frac{(2.2-1)^2}{2}\ \text{Nm}$$

$$= 7680\ \text{Nm}$$

Problem 3.4 Draw S.F. and B.M. diagrams for the simply supported beam shown in Fig. 3.19.
Solution: Let R_a and R_b be the vertical reactions at the supports A and B respectively. For the equilibrium of the beam taking moments of the forces on the beam about the left support, we have,

$$R_b \times 8 = 4 \times 1.5 + 10 \times 4 + 3 \times 6 = 64\ \text{kN/m}$$

$\therefore$ $R_b = \dfrac{64}{8} = 8$ kN

$\therefore$ R_a = total load on the beam – R_b = 17-8 = 9 kN

S.F. between A and C = 9 kN

S.F. between C and D = + 9 -4 = + 5 kN

S.F. between D and E = +9 – 4 – 10 = - 5kN

or alternatively = - 8+3 = - 5 kN

S.F. between E and b = +9 -4 – 10 -3 = - 8 kN

or alternatively = $-R_b$ = - 8 kN

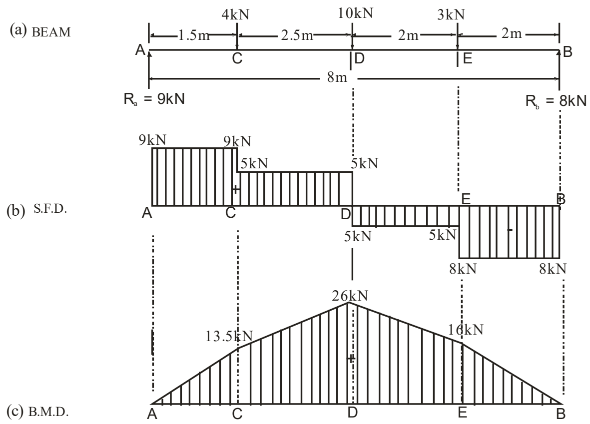

Fig. 3.19

B.M. at A =0

B.M. at C = +9 $\times$ 1.5 = +13.5 kNm (sagging)

B.M. at D = +9$\times$4 - 4$\times$2.5 = +26 kNm (Sagging)

B.M. at E = + 8$\times$2 = + 16 kNm (Sagging)

It may be observed from the S.F. and B.M. diagrams that the maximum B.M. occurs at D where the S.F. change its sign. SFD and BMD are shown in Fig. 3.19.

Problem 3.5 A beam AB 10 meters long has supports at its ends A and B. It carries a point load of 10 kN at 3 meters from A and a point load of 5 kN at 7 metres from A and a uniformly distributed load of 1 kN per meter between the point loads. Draw SF and BM diagrams for the beam.

Solution:

Reactions :

Since the loading is symmetrical reaction at each support equals half the total load

$$\therefore \quad R_a = R_b = \frac{10+10+2\times4}{2} = 14 \text{ kN}$$

S.F. analysis

S.F. at any section in AD = + 14 kN

S.F. just on RHS of D = +14 -10 = + 4 kN

S.F. just on LHS of E = - 14 + 10 = -4 kN

S.F. at any section in EB = - 14 kN

S.F. at centre C = 0

B.M. analysis

B.M. at A = BM at B = 0

B.M. at D = BM at E = + 14 $\times$ 3 = + 42 kNm

$$\text{B.M. at C} = 14 \times 5 - 10 \times 2 - 2 \times \frac{2^2}{2} = 70 - 20 - 4 = +46 \text{ kNm}$$

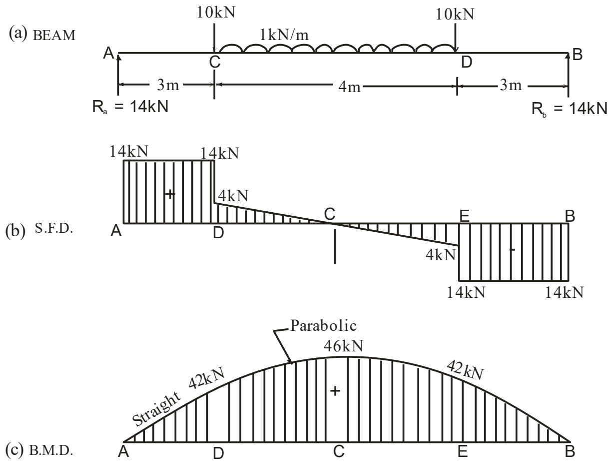

Fig. 3.20

(Maximum bending moment)
The B.M. diagram is linear for the parts AD and BE and is parabolic for the part DE. SFD and BMD are shown in Fig. 3.20.

Problem 3.6 A simply supported beam is loaded as shown in Fig. 3.21. Draw the S.F. amd B.M. diagrams and find the value of maximum bending moment.

Solution: Let R_A and R_B be the reactions at the supports A and B respectively.

$$R_A + R_B = 30 \times 2 + 120 + 50 \times 4 = 380 \text{ kN} \quad \text{.....}(i)$$

Taking moment about A

$$R_B \times 8 = 30 \times 2 \times \left(\frac{2}{2} + 6\right) + 120 \times 6 + 50 \times 4 \times \frac{4}{2} = 1540$$

$$\therefore \quad R_B = \frac{1540}{8} = 192.5 \text{ kN}$$

Putting the value of R_B in Equation (i) we get R_A = 380 – 192.5 = 187.5 kN

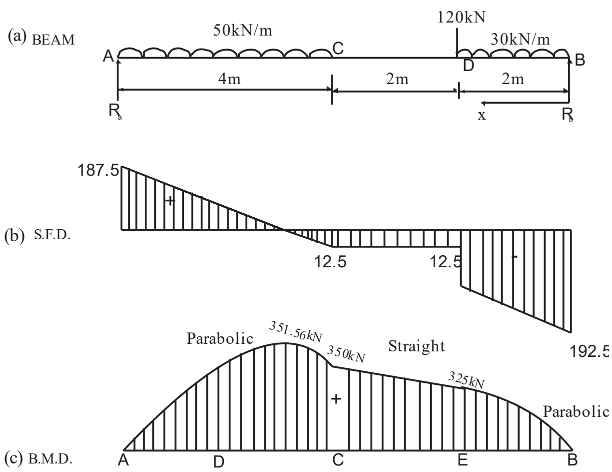

Fig. 3.21

S.F. at B = 192.5 kN (-) ve
S.F. at C = 192.5 – 30 $\times$ 2 – 120 = 12.5 kN (-) ve
S.F. at D = 12.5 kN (-) ve

S.F. at A = 187.5 kN (+) ve

The S.F. diagram is shown in Fig. 3.21 There is a point between A and D where S.F. is zero. Let the point be E and BE = x

Now S.F. at $\quad E = -192.5 + 30 \times 2 + 120 + 50\,(x - 4\,)$

$$0 = -12.5 + 50\,x + 200$$

or $\quad 50x = 212.5$

$\therefore \quad x = 4.25$ m

B.M. at A = 0

B.M. at B = 0

$$\text{B.M. at C} = 192.5 \times 2 - 30 \times 2 \times \frac{2}{2} = 325 \text{ kN.m}$$

$$\text{B.M. at D} = 192.5 \times 4 - 30 \times 2 \times \left(\frac{2}{2} + 2\right) - 120 \times 2 = 350 \text{ kN.m}$$

$$\text{B.M. at E} = 192.5 \times 4.25 - 30 \times 2 \times (\,4.25 - 1\,) - 120 \times 2.25$$

$$- 50 \times 0.25 \times \frac{0.25}{2} = 351.56 \text{ kN.m}$$

Therefore the maximum bending moment is at E and is equal to 351.56 kN.m.

The B.M. diagram is shown in Fig. 3.21.

Problem 3.7 A simply supported beam ABC with supports at A and B, 6 meters apart and with an overhang BC 2 meters long carries a uniformly distributed load of 30 kN per meter over the whole length as shown in Fig. 3.22 Draw S.F. and B.M. diagrams.

Solution:

Reactions. Taking moments about the end A,

$$R_b \times 6 = 30 \times 8 \times \frac{8}{2}$$

$\therefore \quad R_b = 160$ kN

$\therefore \quad R_a = \text{Total load} - R_b = (30 \times 8) - 160 = 80$ kN

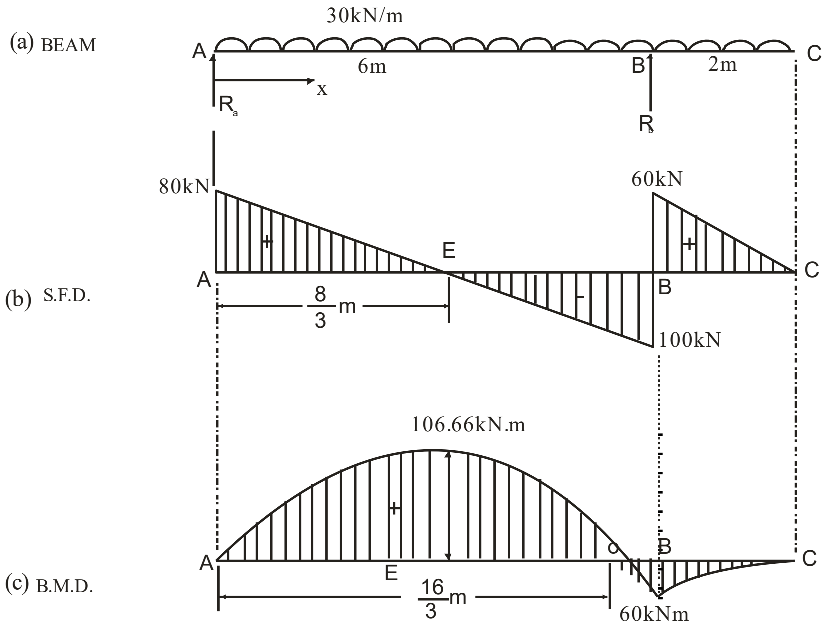

Fig. 3.22

S.F.D.: At any section AB distant x from A,

Shear force $S = 80 - 30x$

At $x = 0$, $S = +80$ kN

At x = 6 m $S = 80 - 30 \times 6 = -100$ kN

Section of zero shear: Equating the general expression for shear force to zero.

$$80 - 30x = 0$$

$$\therefore \quad x = \frac{8}{3} \text{ m} = 2.67 \text{ m}$$

At any section in CB, distant x from C,

Shear force $S = +30x$

At $x = 0$ $S = 0$

At $x = 2$m, $S = 30 \times 2 = +60$ kN

B.M.D.: At any section in AB, distant x from A,

Bending moment $= M = 80x - 30 \times \dfrac{x^2}{2}$

$\therefore$ $M = 80x - 15x^2$

This is a parabolic law.

At x = 0, $M = 0$

At x = 6 m, $M = 80 \times 6 - 15 \times 6^2 = -60$ kNm

Maximum bending moment which occurs at the section of zero shear

$$= M_e = 80 \times \frac{8}{3} - 15\left(\frac{8}{3}\right)^2$$

$$= +\frac{320}{3} \text{ kNm} = +106.66 \text{ kNm}$$

At any section in CB, distant x and C,

Bending moment $= M = -30\dfrac{x^2}{2}$

$\therefore$ $M = -15x^2$. this is a parabolic law.

At x = 0, $M = 0$

At x = 2 m, $M = -15 \times 2^2 = -60$ kNm

Problem 3.8 Draw shear force and bending Moment diagrams for the beam shown in Fig. 3.23.

Solution : Taking moments about A, $R_b \times 10 = 20 \times 14(7-3)$

$\therefore$ $R_b = 112$ KN $\uparrow$ $\therefore$ $R_a = (20 \times 14) - 112 = 168$ KN$\uparrow$

S.F.D.: S.F. at C = 0,

S.F. on LHS of A = $-20 \times 3 = -60$ kN

S.F. on RHS of A = $-60 + 168 = +108$ kN;

S.F. on LHS of B = $168 - 20 \times 13 = -82$ kN

S.F. on RHS of B = $-82 + 112 = +20$ KN ;

S.F. at D = 0

Section of zero shear. Let the shear force be zero at a section distant x from A.

Equating the shear force to zero,

$168 - 20(3+x) = 0$ $\therefore$ $x+3 = 8.4$ $\therefore$ $x = 5.4$ m

B.M.D. BM at C = 0

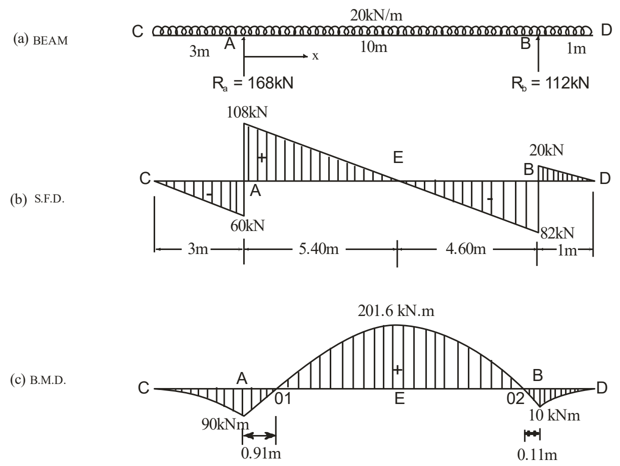

Fig. 3.23

B.M. at A = $\frac{20\times 3^2}{2}$ = - 90 kNm ;

B.M. at B = - $\frac{20\times 1^2}{2}$ = -10 kNm

B.M. at the section of zero shear; i.e. at a distance of 5.40 m from A,

$$M_{max} = 168\times 5.4 - \frac{20\times 8.4^2}{2} = 201.6 \text{ kNm}$$

Points of contra flexure between A and B. Let the B.M. be zero at a distance x from A. Equating the general expression for B.M. to zero , $168x - \frac{20(x+3)^2}{2} = 0$

$\therefore$ $x^2 - 10x + 9 = 0$ solving we get,

$x = 0.91\ m$ and $x = 9.89$ m

Distance of point of contra flexure 0_1 from A = 0.91 *m*

Distance of point of contra flexure 0_2 from B = 10 – 9.89 = 0.11 *m*

Problem 3.9 Calculate the reactions at the supports A and B of the beam shown in Fig. 3.24. Draw bending moment and shearing force diagrams.

Solution: Let reactions at A and B be R_a and R_b respectively.

Taking moments about A , We have , $R_b \times 7 + 1000\times 2$

$= 2500\times 4 + 1500\times 10$

$R_b = \frac{23000}{7} = 3285.71$

$\therefore$ R_a = Total load – R_b = 5000 – 3285.71 = +1714.29

S.F. Calculations

S.F. between D and A = - 1000 N

S.F. between A and C = - 1000 + 1714.29 = + 714.29

S.F. between C and B = + 1500 – 3285.71 = - 1785.71

S.F. between B and E = + 1500 N

B.M. Calculations

B.M. at D = M_d = 0

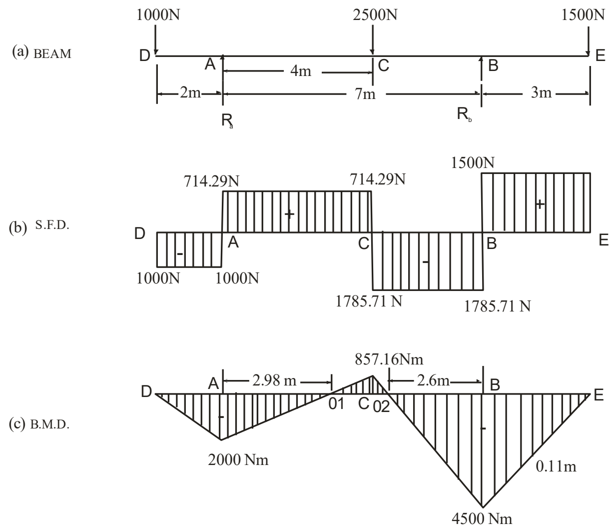

Fig. 3.24

B.M. at A = M_a = - 1000$\times$2 = - 2000 Nm

B.M. at C = M_c = + 1714.29$\times$4-1000$\times$6 Nm

= -857.16

B.M. at B = M_b = - 1500$\times$3 = -4500 Nm

B.M. at E = M_e = 0

There will be two points of contra flexure O_1 and O_2 . One of them lies betweee A and C and the other lies between C and B.

Points of contra flexure between A and C: Let this point be x meters from A. Equating the general expression for B.M. to zero, $1714.29x - 1000(x+2) = 0$

Solving we get, $x = 2.8$ m from A.

Points of contra flexure between C and B: Let this point be x meters from B. Equating the general expression for B.M. to zero, $3285.71x - 1500(x+3) = 0$

Solving we get, $x = 2.52$ m from B.

Problem 3.10 A beam of uniform section 10 meters long carries a uniformly distributed load of 10 kN per meter over the whole length and a concentrated load of 10 kN at the right end. If the beam is freely supported at the left and, find the position of the second support so that maximum bending moment for the beam shall be as small as possible. Find also the maximum bending moment for this case.

Solution: Let the supports be at A and B so that the overhang BC = a meter

$\therefore$ AB = (10 – a) meter

Taking moments about A, $R_b (10 - a) = \dfrac{10\times10\times10}{2} + 10\times10$

$$R_b = \left(\frac{600}{10-a}\right) \text{kN}$$

$$\therefore \quad R_a = (10\times10) + 10 - \frac{600}{10-a} = \frac{500-110a}{10-a} \text{kN}$$

$$\text{B.M. at B} = -10 \times a - 10 \times \frac{a^2}{2} = -5a\,(a+2)$$

Maximum positive B.M. will occure at a section in AB.

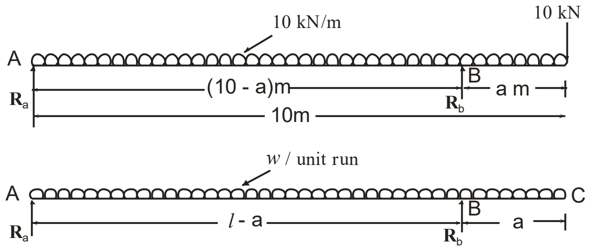

Fig. 3.24

Let this section be x meters from A.

∴ S.F. at this section = 0

∴ $R_a - 10x = 0$

$$\frac{500-110a}{10-a} - 10x = 0$$

∴ $$x = \frac{500-110a}{(10-a)100} = \frac{50-11a}{10-a}m$$

Maximum positive B.M. $$= R_a.x - \frac{wx^2}{2} = \frac{500-110a}{10-a}.\frac{50-11a}{10-a} - \frac{10}{2}\left[\frac{50-11a}{10-a}\right]^2$$

$$= 5\frac{(50-11a)^2}{(10-a)^2}$$

For the condition that the maximum bending moment shall be as possible, the maximum positive bending moment and the hogging moment over the support should be numerically equal.

∴ $$5\frac{(50-11a)^2}{(10-a)^2} = 5a\,(a+2)$$

∴ $$\left[\frac{50-11a}{10-a}\right]^2 = a\,(a+2)$$

Solving by trial and error, we get a = 2.33 m.

Maximum B.M. with the above value of a

$$= 5a\,(a+2) = 5 \times 2.33\,(2.33+2) = 50.44 \text{ kNm}$$

Problem 3.11 A beam AB, 20 meters long supported on two intermediate props 12 meters apart carries a uniformly distributed load 6 kN/m per meter together with concentrated loads of 30 kN at the left end A and 45 kN at the right end B. The supports are so located that the reaction is the same at each support. Determine the position of the props and draw B.M. and S.F. diagrams. Mark the values of the maximum B.M. and S.F.

Solution: Let the left support be at C and the right support be at D

Let AC = a meters

∴ BD = 20 – 12 – a = (8 – a) meters

Let R_1 and R_2 be the reactions at the left and right supports respectively.
Total load on the beam = 35 + 45 + (6 × 20) kN = 200 kN
Since the reactions at the supports are given to be equal, we have

$$R_1 = R_2 = 100 \text{ kN}$$

Taking moments about A, we have ,

$$45 \times 20 + 6 \times 2 \times 10 = 100a + 100\,(12 + a)$$

$$900 + 1200 = 100a + 1200 + 100a$$

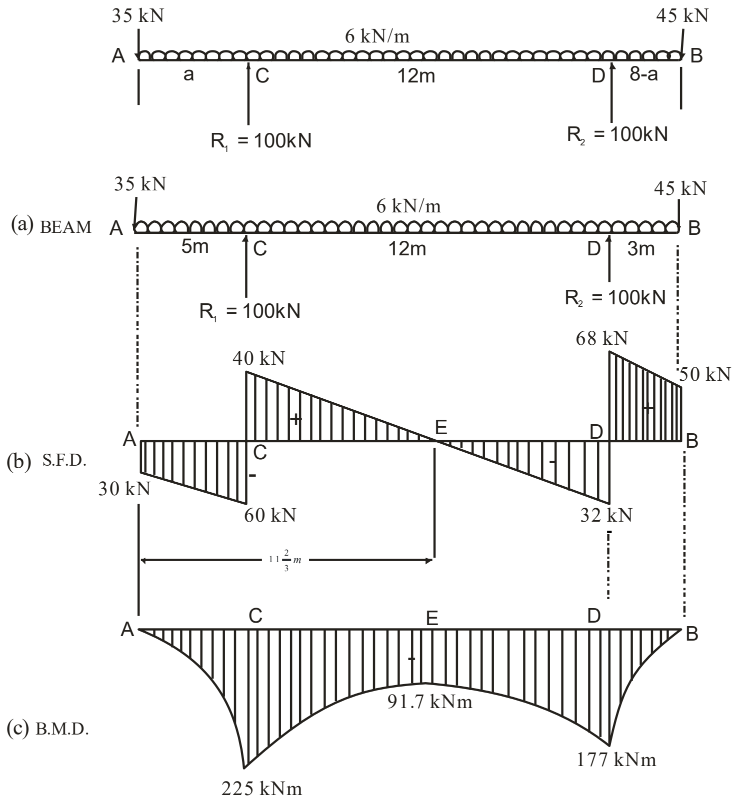

Fig. 3.25

200a = 900

∴ $a = 4.5$ meters

∴ The left support is at 4.5 meters from A and the right support is at 8 – 4.5 = 3.5 meters from B.

Shear force calculations

S.F. just on the right hand side of A	=	- 30 kN
S.F. just on the left hand side of C	=	- 30 -6 × 4.5 = - 57 kN
S.F. just on the right hand side of C	=	- 57 + 100 = + 43 kN
S.F. just on the right hand side of D	=	+ 45 + 6 × 3.5 = 66
S.F. just on the left hand side of D	=	+ 66 – 100
	=	- 34 kN

Let the S.F. be zero at a distance of x meters from A (between the two supports)

Equating the S.F. to zero, we have,

$$100 - 35 - 6x = 0$$

∴ $6x = 65$

$\therefore \qquad x = \dfrac{65}{6} = 10\dfrac{5}{6}$ m

Bending moment calculations

B.M at A = 0

B.M. at C = $M_d = -35 \times 4.5 - 6 \times \dfrac{4.5^2}{2}$ kNm = - 218.25 kNm

B.M. at D = $M_d = -45 \times 3.5 - 6 \times \dfrac{3.5^2}{2}$ kNm = - 194.25 kNm

B.M. at distance of $10\dfrac{5}{6}$ meter from A

$$= M_e = 100 \times 5\frac{5}{6} - 30 \times 10\frac{5}{6} - \frac{6\left(10\frac{5}{6}\right)^2}{2} \text{ kNm} = -93.75 \text{ kNm}$$

Problem 3.12 Calculate the reactions for the beam shown in Fig. 3.26 Construct the bending moment and shear force diagrams. Determine the location of the maximum bending moment and mark it clearly on each of the diagrams.

Solution: Let R_a and R_c be the reactions at the supports *A* and *C* respectively.
Taking moments about *A*, we have

$$R_c \times 5 = 80 \times \frac{3}{2} + 15 \times 7$$

$\therefore \qquad R_c = 45$ kN

$\therefore \qquad R_a = 80 + 15 - 45 = 50$ kN

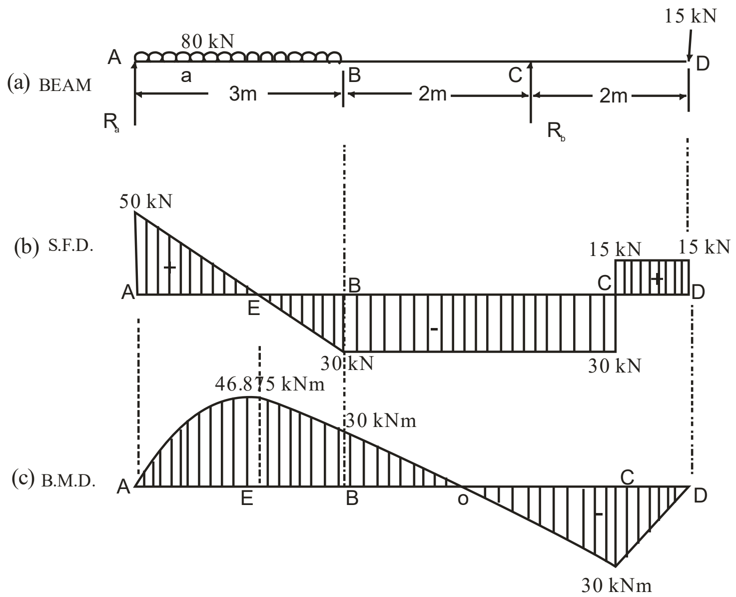

Fig. 3.26

S.F. Calculations

S.F. at A = + 50 kN

S.F. at B = 50 – 80 = - 30 kN

S.F. between B and C = - 30 kN

S.F. between C and D = + 15 kN

Let the S.F. be the zero at x meters from A.

Equating the S.F. to zero, we have,

$$50 - \frac{80}{3}x = 0$$

$$\therefore \quad x = \frac{3}{80} \times 50 = 1.875 \ m \text{ from } A$$

B.M. Calculations

B.M. at A = M_a = 0

B.M. at B = M_b = $50 \times 3 - 80 \times \frac{3}{2}$ kNm = + 30 kNm

B.M. at E *i.e.*, 1.875 meters from A

$$= 50 \times 1.875 - \frac{80}{3}\frac{(1.875)^2}{2} \text{ kNm} = +46.875 \text{ kNm}$$

B.M. at C = M_c = - 15 $\times$ 2 = - 30 kNm

B.M. at D = M_d = 0

Point of contra flexure

There will be a point of contraflexure between B and C.

Let this point of contraflexure be at x meter from C

Equating the B.M. to zero, we get,

$$40x - 15\,(2 + x) = 0$$

$$\therefore \quad x = 1.2 \text{ m}$$

Exercise

3.1 What are the different types of beams?

3.2 What are the sign conventions for shear force and bending moment?

3.3 What do you mean by point of contraflexure?

3.4 What are the different types of load acting on a beam?

3.5 Draw SFD and BMd for a cantilever subjected to udl.

3.6 A cantilever of length 2 m carries a udl of 3 kN/m over a length of 1 m from the fixed end. Draw SFD and BMD.

(Ans. F_{max} = 3kN, M_{max} = -1.5kNm)

3.7 A cantilever of length 4 m carries a udl of 1 kN/m over whole length and a point load of 2 kN at 1 m from the free end. Draw SFD and BMD.

(Ans. F_{max} = 14kN, M_{max} = -14kNm)

3.8 A simply supported beam of length 8 m carries point loads of 4, 10 and 7 kN at a distance of 1.5, 2.5 and 2 m respectively from left end. Draw the SFD and BMD.

(Ans. M_{max} = 90kNm)

3.9 A simply supported beam of length 8 m carries point loads of 4 and 6 kN at a distance of 2 and 4 m respectively from left end. Draw the SFD and BMD.

(Ans. M_{max} = 20kNm)

3.10 A simply supported beam of length 6 m carries point loads of 1, 2 and 3 kN at a distance of 1.5, 3 and 4.5 m respectively from left end. There is a udl of 1.5 kN/m over the entire length. Draw the SFD and BMD.

(Ans. M_{max} = 12.75kNm)